

7. According to the World Health Organisation, the failure of parents to vaccinate their children in 2016 led to the biggest surge in measles cases in Europe for over a decade. Across 53 countries, there were 37 deaths reported from a total of 41 000 cases of measles.

(a) (i) Calculate the death rate for measles per 1000 cases of measles. [1]

Death rate = per 1000 cases of measles

(ii) Suggest a reason why some parents choose not to have their children vaccinated despite the risk to their health. [1]

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(b) Explain why an individual who has contracted measles once is unlikely to contract the disease a second time. [4]

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(c) Typhoid is a serious disease caused by the bacterium *Salmonella typhi*. Individuals who travel to the Indian subcontinent, Africa, South America and southeast Asia are advised to be vaccinated. The typhoid vaccine contains fragments of *S. typhi*.

(i) Explain why an individual receiving the typhoid vaccine will not develop the disease even though it contains fragments of *S. typhi*. [1]

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(ii) Explain why the typhoid vaccine does not prevent fevers or infections caused by other species of *Salmonella* bacteria. [1]

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(d) State how an increased understanding of the human genome can be important for medicine. [1]

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8. Diabetes is a condition where the concentration of glucose in the blood can become too high. Over time, this may cause complications including damage to blood vessels and nerves.

(a) (i) Apart from high blood glucose levels, state **one** other symptom of diabetes. [1]

(ii) Insulin is essential in the control of blood glucose. With reference to insulin, explain the difference between Type 1 and Type 2 diabetes. [2]

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(iii) Apart from insulin injections, describe **one** other possible treatment for people with diabetes. [1]

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(b) Explain how a negative feedback mechanism raises blood glucose levels in a healthy individual when their glucose levels are decreasing. [3]

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- (c) **Table 8.1** and **Table 8.2** show information on the ethnic categories of 26 907 children and young people up to the age of 24 years in Wales and England in 2016/17 who were diagnosed with either Type 1 or Type 2 diabetes.

Table 8.3 shows the ethnic categories of **all** children and young people in Wales and England.

Table 8.1 – Ethnic categories of children and young people with **Type 1 diabetes, 2016/17**

Ethnic category	Number	Percentage of the sample of children who had Type 1 diabetes (%)
White	22 111	84.0
Mixed	699	2.7
Asian	2 136	8.1
Black	956	3.6
Other	421	1.6
Total number of children with Type 1 diabetes =	26 323	

Table 8.2 – Ethnic categories of children and young people with **Type 2 diabetes, 2016/17**

Ethnic category	Number	Percentage of the sample of children who had Type 2 diabetes (%)
White	246	42.1
Mixed	29	5.0
Asian	212	36.3
Black	79	13.5
Other	18	3.1
Total number of children with Type 2 diabetes =	584	



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Table 8.3 – Ethnic categories of **all children and young people in Wales and England**

Ethnic category	Percentage of population of all children and young people in Wales and England (%)
White	86.0
Mixed	2.2
Asian	7.5
Black	3.3
Other	1.0

- (i) Calculate the ratio of the total number of children and young people with Type 1 diabetes against the total number with Type 2 diabetes. [1]

Type 1 : Type 2

- (ii) Use **Table 8.1**, **Table 8.2** and **Table 8.3** to state **one** conclusion that can be drawn on the effect of ethnicity on the percentage of children and young people affected by:

I. Type 1 diabetes; [1]

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II. Type 2 diabetes. [1]

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- (d) A politician looking at these data concluded that the number of people in the whole population of Wales and England with Type 1 diabetes is far higher than the number with Type 2 diabetes. State why this is not a valid conclusion. [1]

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